Melvin Khakabo

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Professional Summary

Research Scientist and AI Education Leader with a proven track record in developing machine learning models and building comprehensive AI educational programs. Combines deep technical expertise in Python, R, and data analysis with strong leadership skills to drive innovation and skill development across diverse audiences, from corporate clients to young students. Proven ability to manage full project lifecycles, lead cross-functional teams, and deliver data-driven results.

Professional Experience

Head of AI & STEM Programs | Learning Sprouts School of AI, Nairobi
June 2024 – Present

- Defined the strategic vision and launched 5+ new AI course streams for students and corporate clients.
- Mentored and managed a team of 10+ instructors, improving teaching efficacy by 30% through targeted coaching.
- Researched emerging trends to develop 15+ project-based curricula, leveraging analytics to achieve a 25% increase in course completion rates.
- Pitched and secured AI training contracts with 5+ major Kenyan corporations by effectively communicating ROI to non-technical decision-makers.
- Managed the end-to-end program lifecycle for events serving 500+ participants, optimizing resources to reduce operational overhead by 15%.

Data Scientist | HackBio
August 2024 – May 2025

- Utilized R and Python for data analysis and visualization on complex cancer datasets, ensuring 100% streamlined data processing.
- Employed machine learning techniques (Random Forest, K-Nearest Neighbors) in the full model lifecycle, achieving 95% prediction accuracy.
- Led cross-functional teams using JIRA to manage 100+ software development tasks, ensuring 90% of project milestones were delivered on time.
- Communicated project updates via Microsoft Teams and Slack, managing 8+ project timelines and minimizing delays by 25%.

Lead Research Coordinator | OCENA
Seasonal Contract

- Developed and conducted 500 comprehensive WASH surveys for CSOs and government organizations using the Kobo Collect tool in collaboration with KEWASNET.
- Implemented robust data handling procedures built on 100% data quality and accuracy to maintain the integrity and confidentiality of all collected data.
- Analyzed survey data using MYSQL to develop valuable insights into the national WASH landscape, which culminated in a final report presented to various stakeholders.

Technical Research Associate | Computer Science Academy Africa
July 2023 - August 2024

- Conducted intensive market research on programming awareness among university students across Africa, ensuring 100% completion rate across participating countries.
- Coordinated a 100-person team to perform data collection through surveys and interviews, streamlining communication and conflict resolution.
- Analyzed collected data to develop and present a report that provides tangible insights on programming opportunities for African students.

Technical Skills

- **Programming Languages:** Python, R, SQL, JavaScript, HTML, CSS, Dart, TypeScript, CSS
- **Frameworks & Technologies:** Django, Node.js, Express.js, React, REST API, GraphQL API
- **Databases & Tools:** PostgreSQL, SQL Server, MySQL, SQLite, Git
- **ML & Data Science:** Machine Learning, Data Analysis, Data Visualization, CI/CD (GitHub Actions, BitBucket Pipelines)
- **Project Management:** Jira, Trello, Slack, Microsoft Teams, Agile Methodologies

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Projects

SALAMA – Survivor Support Web app | *Full-Stack Developer*

- Built a comprehensive web app using Node.js, Express, and ReactJS for an AI-powered mobile platform disguised as a normal utility app that helps survivors safely document abuse, access support, and get help without being detected.
- Created a dynamic user interface for streamlined evidence encryption, a 24/7 AI companion for 24/7 trauma-informed guidance, and emergency SOS alerts to trusted contacts.

ReproGuard | *Full-Stack Developer*

- Developed a health-tech application using Node.js, Express, and ReactJS to provide anonymous reproductive health advice.
- Engineered the backend with SQLite, implementing database replication for high availability and data integrity.
- Incorporated advanced API resilience techniques, including rate limiting and retries to maintain service reliability.

Education

Bachelor of Science in Biomedical Science and Technology
Egerton University · Nakuru, Kenya · 2023

Certifications

- Cybersecurity | ALX (2025)
- AWS Cloud Computing | AWS (2025)
- Data Science and Machine Learning | HackBio (2024)
- Software Development | Power Learn Project (2024)
- Aspire Leadership Program | Harvard (2023)
- AI Career Essentials + AI Toolkit | ALX Africa (2023)